

Progress Report and Research Plan

Alpha Models Across Market Regimes

Adhish Rajan

Roshan Patel

Jean Sainte-Beuve

Brandon Li

15.S06

October 2026

Abstract

Return predictors are unstable: a signal that works in one environment can fail or reverse in another. We study whether a mixture-of-experts (MoE) model, in which economically defined “domain experts” are combined by a gate that conditions on the current state, can adapt to this instability better than pooled or equal-weighted combinations. We define experts using the 13 characteristic themes of [Jensen et al. \(2023\)](#) and build the US stock panel from the JKP database (largest 2,000 stocks, 1990–2025). Preliminary evidence shows that (i) theme payoffs are highly unstable over time, (ii) an equal-weighted combination of 13 unfitted theme experts is already a strong benchmark (decile long-short Sharpe 0.90), and (iii) expert skill varies substantially with *stock-level* states such as volatility, size and past returns, while *market-level* regimes offer far fewer independent observations to learn from. We describe a common training and evaluation protocol for fitted experts and the planned gating models.

1 Motivation and research questions

Financial predictors are noisy and non-stationary. Characteristic-based anomalies weaken after publication, crash in specific episodes (e.g. momentum in 2009), and appear to work better for some types of stocks than others. A natural response is to let the *combination* of signals depend on the state of the world. A mixture-of-experts model ([Jacobs et al., 1991](#); [Shazeer et al., 2017](#)) does exactly this: several expert models each produce a forecast, and a gating network assigns state-dependent weights to the experts.

Our design is motivated in part by a practitioner example presented by a guest speaker from Trexquant, in which four alpha signals (news, fundamental, price–volume and reversion) are combined using a gate that conditions on roughly 20 *stock-specific* state variables (fundamental ratios, momentum, recent returns and volatility, days to earnings, news sentiment). A key distinction for us is between two kinds of conditioning information:

- **Market-level regimes** (e.g. up/down markets, high/low volatility) are the same for every stock in a given month. With roughly 400 months of data and highly persistent states, a gate learns from only a handful of independent regime episodes.
- **Stock-level states** (e.g. a stock’s size, volatility, liquidity or past return) vary across the cross-section every month, so a gate learns from hundreds of thousands of stock-month observations.

Research questions.

1. Does grouping signals by economic theme produce experts that genuinely specialize across stocks and over time, relative to a single pooled model and simple forecast averaging?
2. Can a soft or sparse gate use *ex ante* state information, at the stock level and at the market level, to weight experts out of sample? Are the learned routing weights stable and economically interpretable?
3. How does the MoE compare with equal weighting, a pooled machine-learning model, Bayesian model averaging (Avramov, 2002) and regime-switching benchmarks (Ang and Timmermann, 2012) in predictive accuracy, portfolio performance, turnover and robustness?

2 Related literature

Factor zoo and characteristic themes. Jensen et al. (2023) construct 153 characteristic-based factors, show that most replicate, and group them into 13 themes by clustering factor returns. These themes are our expert definitions. Gu et al. (2020) show that flexible machine-learning models applied to a large set of characteristics substantially improve out-of-sample return prediction; their pooled models are the natural “no structure” benchmark. Kelly et al. (2019) let factor loadings depend linearly on stock characteristics, a linear analogue of stock-level conditioning.

State-dependent predictability. Cooper et al. (2004) show that momentum profits depend on the prior market state. Nagel (2012) links short-term reversal profits to liquidity provision, which is more valuable in turbulent markets. Heston and Sadka (2008) document seasonality in the cross-section.

Forecast combination and model uncertainty. Equal-weighted combinations are notoriously hard to beat (DeMiguel et al., 2009; Rapach et al., 2010). Avramov (2002) applies Bayesian model averaging to return predictability. Ang and Timmermann (2012) review regime-switching models in finance.

Neural mixture-of-experts for stock ranking. Kim and Song (2026) combine vector-quantized latent factors, prior factors and a code-conditioned MoE. Their S&P 500 rank IC of 0.014 illustrates how weak cross-sectional signals are among large stocks. We deliberately use a simpler and more interpretable design.

Common task framework. Hellum et al. (2025) propose a common task framework for finding the stochastic discount factor on JKP data, with a public leaderboard and benchmark models. Its 1990–2023 test period and rules (no external data, monthly rebalancing) inform our design.

3 Data

3.1 Source and sample

We use the JKP global stock panel (`contrib.global_factor` on WRDS), restricted to US stocks (`excntry = USA`) and the four standard JKP screens: common stocks, main exchanges, the primary security of each firm, and the CRSP observation where CRSP and Compustat overlap. The panel is at the stock-month level and contains the 153 JKP characteristics, market capitalization

(`me`), a size group based on NYSE breakpoints (`size_grp`), and the next-month excess return (`ret_exc_lead1m`). Our sample starts in January 1990.

Timing. Characteristics are measured as known at the end of month t (JKP lag accounting data before use), and the prediction target is the excess return earned over month $t + 1$. All rankings, universe membership and portfolio weights use month- t information only.

3.2 Universe

Each month we keep the 2,000 largest stocks by market capitalization. Figure 1 compares this universe with the full set of US common stocks and with the “mega + large” size groups (stocks above the NYSE median). The top-2,000 universe captures 93–99% of US market capitalization throughout, while providing roughly twice the cross-section of the mega + large definition. A fixed count also avoids the drift in universe size that a fixed nominal dollar cutoff would create over 35 years.

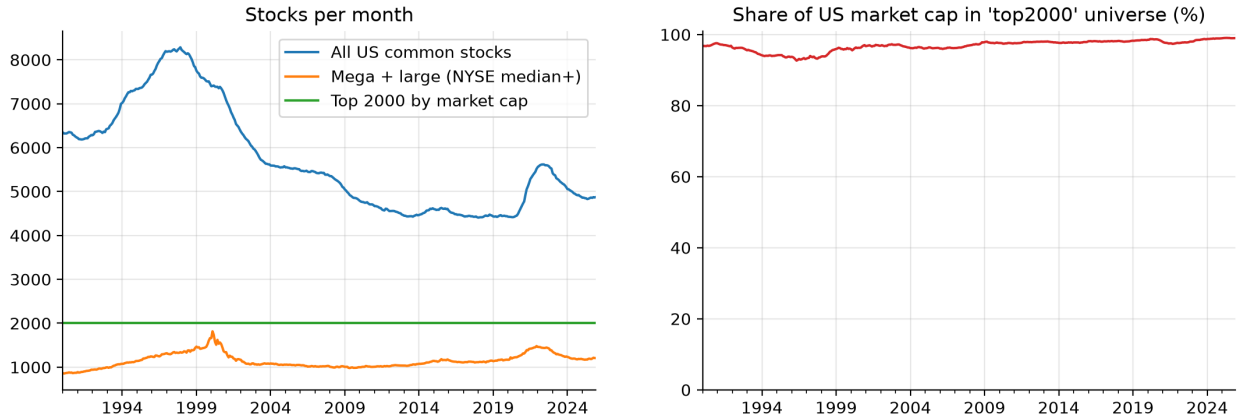


Figure 1: Universe size. Left: number of stocks per month. Right: share of total US market capitalization in the top-2,000 universe.

3.3 Characteristics and themes

The 153 characteristics are assigned to 13 themes using JKP’s published cluster labels (Table 1). Each characteristic also carries a JKP *direction* (± 1): the sign of its documented relation with future returns, which we use so that “higher” always means “higher expected return.”

Table 1: Number of characteristics per JKP theme.

Theme	# chars	Theme	# chars
Investment	22	Profitability	11
Value	18	Momentum	8
Low Risk	18	Debt Issuance	7
Quality	17	Accruals	6
Seasonality	12	Short-Term Reversal	6
Profit Growth	12	Size	5
Low Leverage	11		
Total: 153 characteristics			

Missing data. Figure 2 shows the share of missing characteristics by theme and year. Seasonality is the most incomplete (20–30%), because its signals require many years of return history that young firms lack. Accruals and Low Leverage are thin before 2005. A broad increase in 2021 coincides with the wave of new listings (IPOs and SPACs) visible in Figure 1. Missing values are set to the cross-sectional median (zero after centring), which means that sparse themes carry less information for young firms.

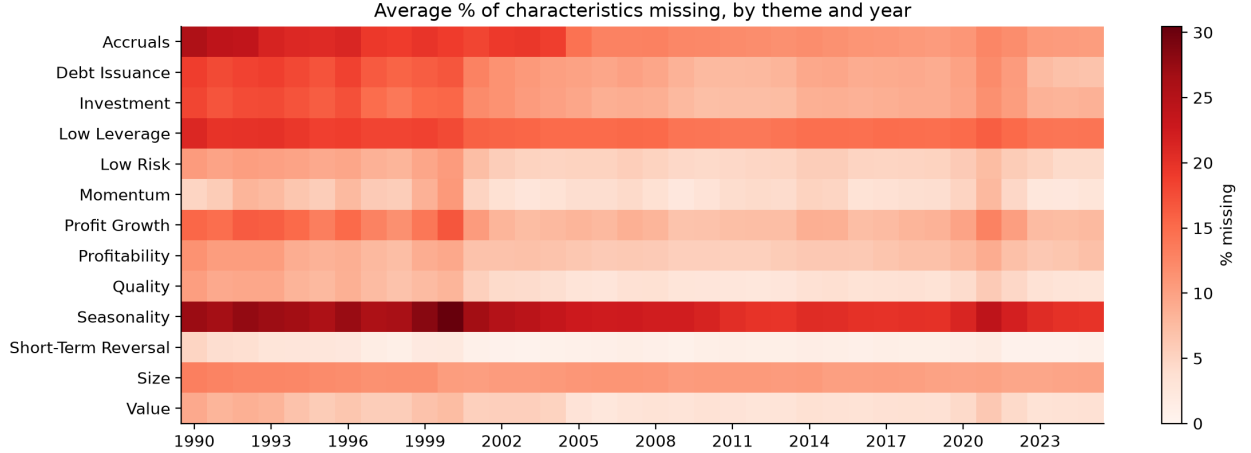


Figure 2: Average percentage of characteristics missing, by theme and year (top-2,000 universe).

4 Preliminary analysis

4.1 Signal construction

Each month t , every characteristic is ranked across the stocks in the universe, converted to a percentile, centred to $[-0.5, 0.5]$, multiplied by its JKP direction, and missing values are set to 0. We denote the resulting signed rank of characteristic j for stock i by $s_{ij,t}$.

4.2 Rebuilding the JKP theme factors

For each characteristic we form a tercile long-short portfolio: long the top third of stocks by $s_{ij,t}$, short the bottom third, using value weights capped at the 80th percentile of market capitalization, held for one month. A theme factor is the equal-weighted average of its characteristics' long-short returns, following [Jensen et al. \(2023\)](#). These rebuilt factors serve both as a check of our data pipeline (to be compared with the published JKP US theme factors) and as a description of how each theme has performed.

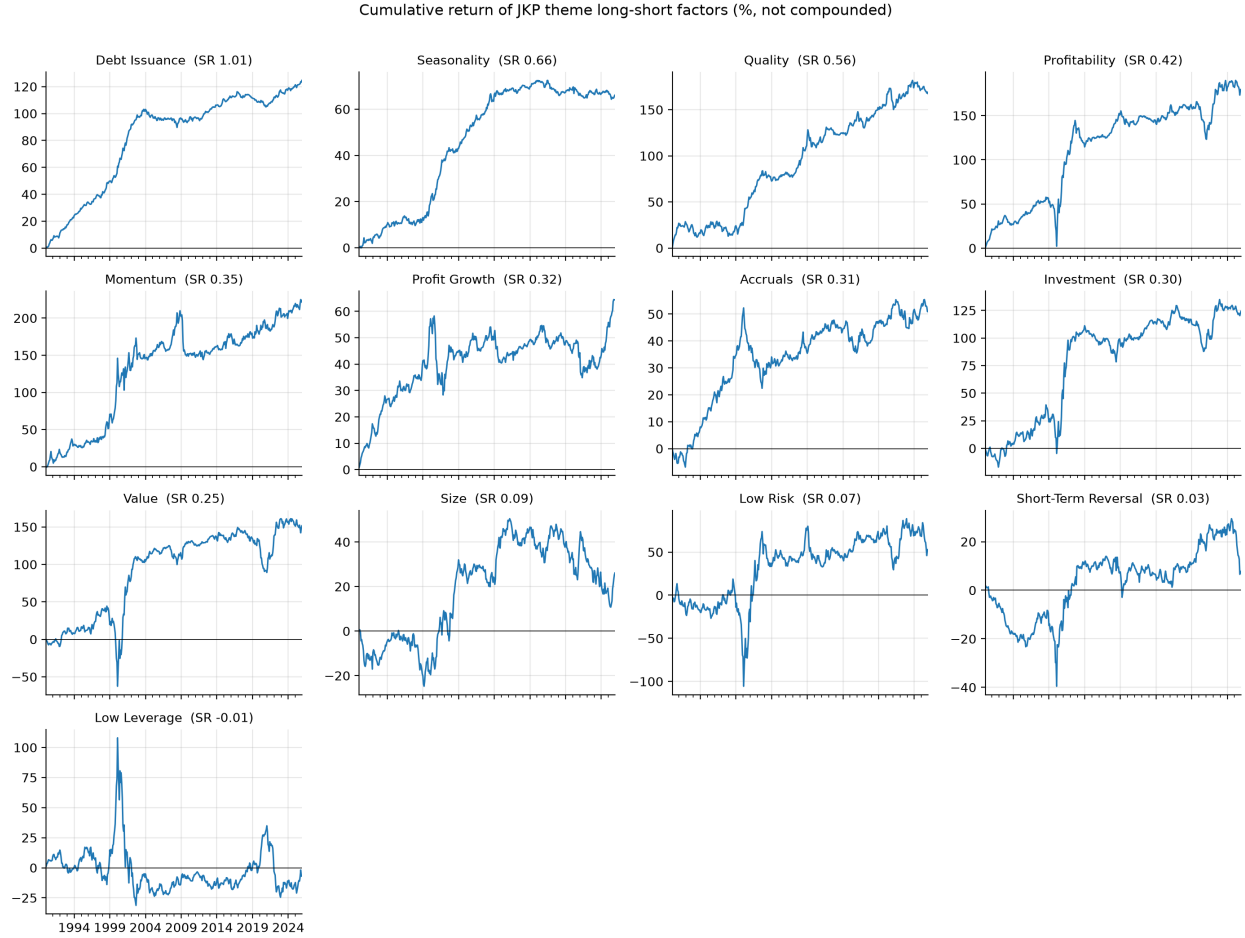


Figure 3: Cumulative return of the 13 theme long-short factors, 1990–2025 (summed monthly returns, %). Titles report full-sample annualized Sharpe ratios.

Figure 3 shows that among the largest 2,000 stocks, only a few themes (Debt Issuance, Seasonality, Quality) earn solidly positive premia over the full period, while Size, Low Risk, Short-Term Reversal and Low Leverage are close to zero. This is consistent with anomalies being weaker among large stocks and after publication. Familiar episodes are clearly visible: the 1999–2001 technology bubble and crash (Value and Low Risk draw down sharply, mirrored by Low Leverage), the 2009 momentum crash, the 2007–2020 stagnation of Value and its 2021–22 recovery, and the flattening of Seasonality after its publication (Heston and Sadka, 2008).

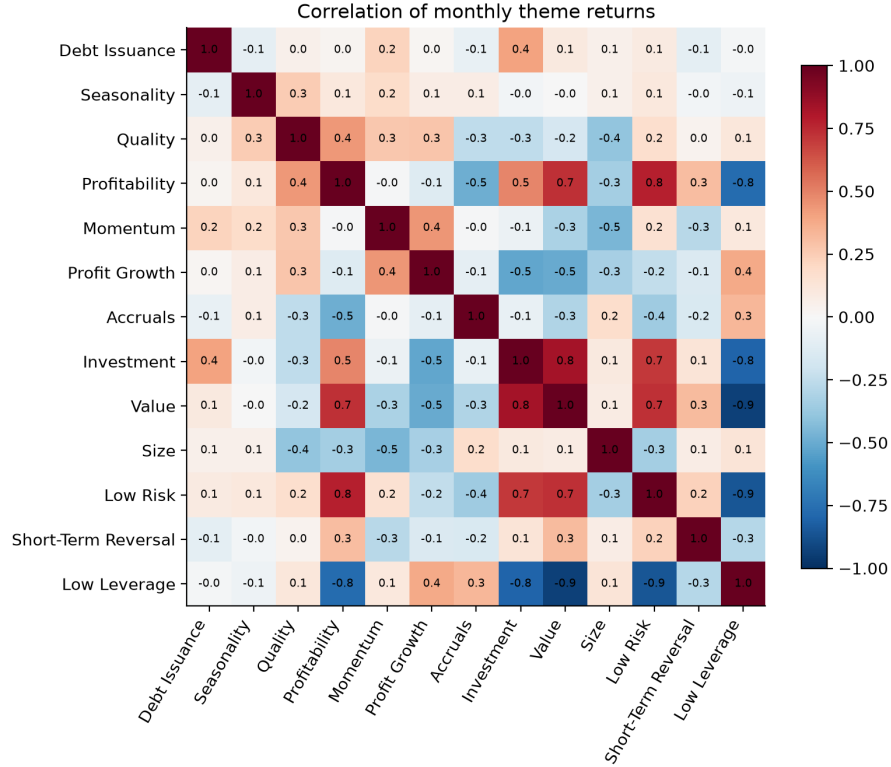


Figure 4: Correlation of monthly theme factor returns.

Figure 4 shows that the themes are far from independent. Value, Investment, Profitability and Low Risk form a positively correlated block (pairwise correlations of 0.5–0.8), and Low Leverage is close to their mirror image (correlations of -0.8 to -0.9). The 13 experts therefore represent fewer independent bets, perhaps six to eight, which matters for interpreting gate weights.

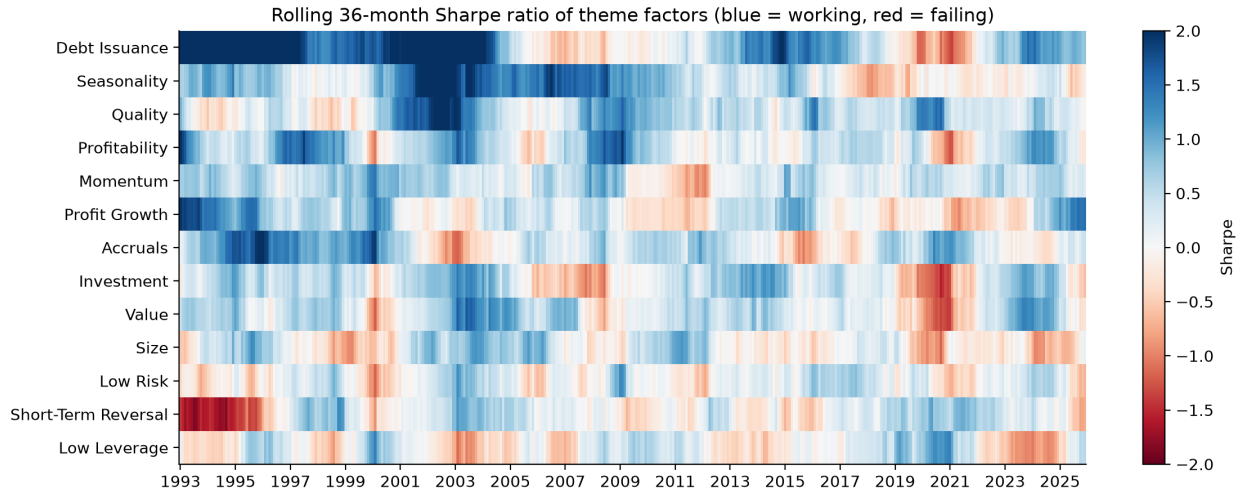


Figure 5: Rolling 36-month Sharpe ratio of each theme factor. Blue: the theme is working; red: it is failing.

Figure 5 is the central motivation for gating: no theme earns its premium steadily. Every theme experiences multi-year periods of negative performance, and several fail at the same time, notably

around 2000–01 and 2020–21.

4.3 Naive theme experts

As a baseline before any model fitting, we define a *naive expert* for each theme k : the score of stock i in month t is the average signed rank of the theme’s characteristics, re-ranked within the month,

$$e_{i,t}^{(k)} = \text{rank}_t \left(\frac{1}{|J_k|} \sum_{j \in J_k} s_{ij,t} \right) - 0.5. \quad (1)$$

The equal-weighted combination (EQWT) averages the 13 expert scores and re-ranks. Each score is evaluated by its monthly rank information coefficient (IC; the Spearman correlation between scores and next-month returns across stocks) and by a decile long-short portfolio (long the top 10% of stocks by score, short the bottom 10%, capped value weights).

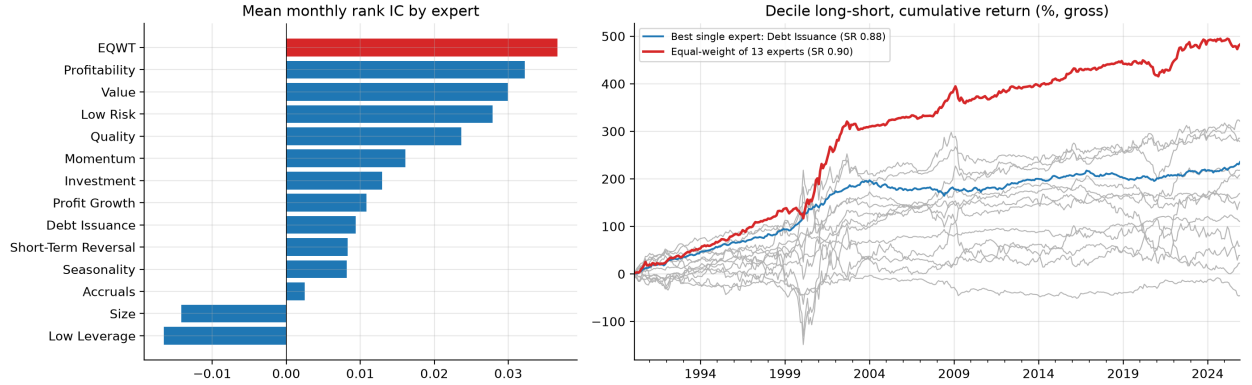


Figure 6: Naive expert baselines. Left: mean monthly rank IC. Right: cumulative return of decile long-short portfolios (gross of costs); grey lines are individual experts.

Table 2: Naive expert decile long-short portfolios, 1990–2025 (capped value-weighted, gross of costs).

Expert	Ann. return (%)	Ann. vol. (%)	Sharpe	t -stat
EQWT (13 experts)	13.44	15.01	0.90	5.37
Debt Issuance	6.56	7.48	0.88	5.25
Quality	7.75	13.27	0.58	3.50
Seasonality	4.72	9.40	0.50	3.01
Profit Growth	6.08	13.52	0.45	2.69
Profitability	7.95	19.04	0.42	2.50
Accruals	3.06	8.23	0.37	2.23
Momentum	8.93	24.70	0.36	2.17
Investment	5.81	18.32	0.32	1.90
Value	4.23	27.61	0.15	0.92
Low Leverage	1.64	26.86	0.06	0.37
Size	0.64	12.33	0.05	0.31
Low Risk	1.08	32.19	0.03	0.20
Short-Term Reversal	-1.27	10.05	-0.13	-0.76

Three findings stand out. First, EQWT has the highest IC (about 0.036) and the highest Sharpe ratio (0.90), and its cumulative return is much smoother than that of the best single expert: equal

weighting is a strong benchmark that any gate must beat. Second, IC and portfolio performance can disagree. Low Risk ranks stocks well on average (IC ≈ 0.028) but its extreme deciles crashed in 2000, leaving its long-short Sharpe near zero. Third, the Size and Low Leverage experts have negative ICs in this universe (among the largest 2,000 stocks, larger firms outperformed), so the naive EQWT includes two experts pointing the wrong way. Fitted experts learn their own signs and should improve on this.

4.4 Which expert, when? Conditional skill

Stock-level states. Each month we split stocks into terciles of four state variables (size, 21-day volatility, 126-day dollar volume, and 12-month past return) and compute each expert’s IC within the low and high terciles (Figure 7). Expert skill varies substantially with the state of the stock. The Momentum expert has an IC of 0.035 among high-volatility stocks but -0.004 among low-volatility stocks. The Value expert has an IC of 0.043 among past losers but 0.014 among past winners. The Low Risk expert has an IC of 0.037 among the smaller stocks in the universe and 0.008 among the largest. Importantly, the *ranking* of experts also changes: Quality and Momentum lead among high-volatility stocks, while Profitability and Short-Term Reversal lead among low-volatility stocks. This is precisely the pattern a stock-level gate can exploit. Some of these effects are partly mechanical (the Momentum expert and the past-return state share inputs), and formal significance tests are still to be added.

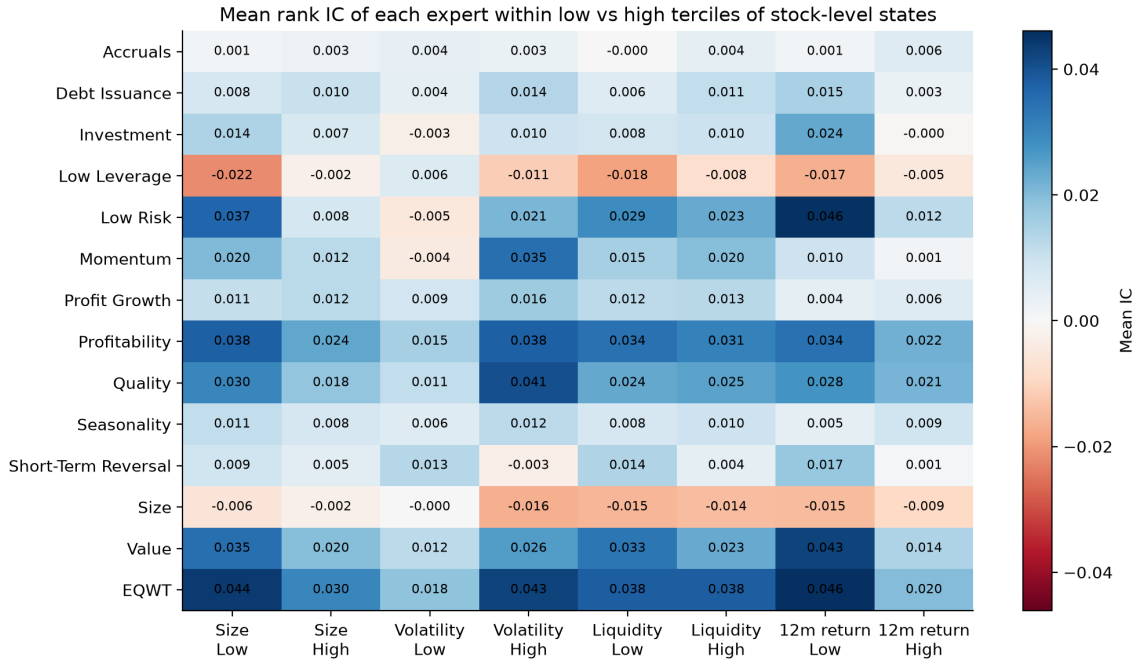


Figure 7: Mean monthly rank IC of each expert within the low and high terciles of four stock-level states.

Market-level states. We construct two market states using only the panel itself: an up/down market indicator based on the trailing 12-month value-weighted universe return (using only returns realized by month t), and a calm/volatile indicator comparing the cross-sectional median 21-day volatility with its own expanding historical median (Figure 8). Several classic results appear: Momentum fails after down markets (IC -0.019 versus 0.026 after up markets), consistent with Cooper et al. (2004); Short-Term Reversal and Value work better in down and volatile markets,

consistent with Nagel (2012). However, there are only 97 down-market months, and because they come from overlapping 12-month windows they amount to perhaps eight to ten distinct episodes. Meanwhile, EQWT is stable across market states (IC 0.033–0.041). A market-level gate therefore faces a strong benchmark with very few independent observations, which motivates our emphasis on stock-level states.

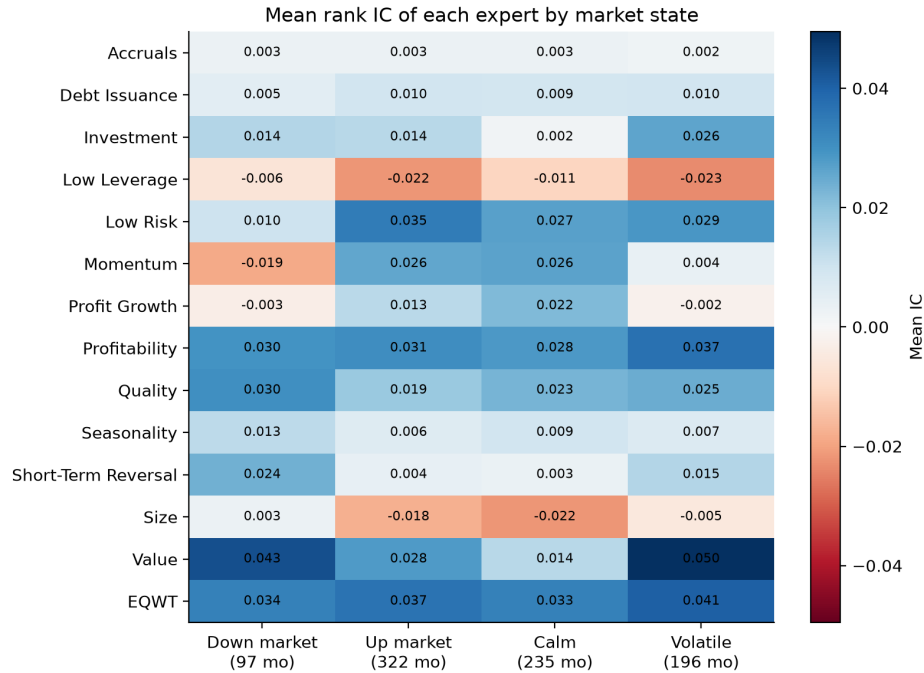


Figure 8: Mean monthly rank IC of each expert by market state; the number of months in each state is shown in parentheses.

5 Research design

To make results comparable across team members, all expert models follow a shared written protocol (RESEARCH_STANDARDS.md in our repository). Its main elements are summarized below.

5.1 Sample splits

Table 3: Fixed project timeline (formation months).

Period	Role
1990–2000	Initial training data for experts
2001–2005	Initial validation block for experts
2006–2015	Expert out-of-sample evaluation and comparison; gate training (2006–2012) and gate validation (2013–2015)
2016–2020	Gate test; not examined during expert development
2021–latest	Final holdout, evaluated once with all design choices frozen

5.2 Expert training

Each expert uses only its own theme’s characteristics (signed, centred ranks, with an optional missing-data indicator). Models are refit once a year on an expanding window. For prediction year Y :

1. Use only formation months up to November of $Y - 1$, whose next-month returns are known by the end of $Y - 1$.
2. Train on 1990 through December of $Y - 6$; validate on January of $Y - 5$ through November of $Y - 1$.
3. Choose hyperparameters by mean monthly rank IC on the validation block (at most 20 configurations, identical across themes and years), refit on train plus validation, and predict all months of year Y .

The training target is the next-month excess return, winsorized within each month at the 1st and 99th percentiles (evaluation uses raw returns). Every expert is reported alongside two baselines: the naive average-rank expert and a ridge regression on the same features.

5.3 Evaluation

All models are evaluated identically:

- **Rank IC:** monthly Spearman correlation between predictions and next-month returns; mean, t -statistic, annualized IC information ratio and share of positive months.
- **Portfolios** (evaluation only): decile (top/bottom 10%) and tercile (top/bottom third) long-short portfolios, each with capped value weights and equal weights, rebalanced monthly.
- **Turnover and costs:** drift-adjusted turnover; Sharpe ratio net of 10 basis points per dollar traded; break-even cost.
- **Out-of-sample R^2** relative to a zero forecast, as in [Gu et al. \(2020\)](#).

5.4 Gating models

Each month the gate receives, for every stock, the 13 expert predictions standardized cross-sectionally (z -scores, clipped at ± 3) together with state variables, and outputs expert weights $g_k(\mathbf{z}_{i,t})$ and a combined forecast

$$\hat{r}_{i,t+1} = \sum_{k=1}^{13} g_k(\mathbf{z}_{i,t}) \hat{r}_{i,t+1}^{(k)}, \quad \sum_k g_k = 1. \quad (2)$$

We will compare:

- gates conditioned on stock-level states only, market-level states only, and both;
- soft (softmax) routing versus sparse top- k routing;
- benchmarks: EQWT of fitted experts, a pooled model on all 153 characteristics, Bayesian model averaging, and a regime-switching combination.

The gate is trained only on the experts’ out-of-sample predictions, so that it cannot learn to favour experts that overfit their own training data. We will also examine whether routing weights are stable over time and economically interpretable.

6 Next steps and open issues

Next steps.

1. Validate the rebuilt theme factors against the published JKP US theme factors.
2. Build a shared code module (data, features, annual refits, evaluation) so every expert uses identical machinery.
3. Train fitted experts (ridge, then gradient-boosted trees) for all 13 themes and compare them with the naive baselines on 2006–2015.
4. Train and evaluate the gating models on the saved out-of-sample expert predictions.
5. Add significance tests for conditional ICs and for differences between models.

Open issues.

- **Redundant experts.** Several themes are highly correlated; we may merge some or report effective numbers of experts.
- **Overlap between states and experts.** Size, Low Risk and Momentum experts contain variables that are also natural gate states; we must decide whether such variables enter the gate, the experts, or both.
- **Mild look-ahead in the setup.** JKP’s directions and theme clusters were derived partly from data after 1990. Fitted experts learn their own signs, which removes the first issue.
- **Weak signals in large stocks.** Restricting to the top 2,000 stocks makes the prediction problem harder but the results more realistic and tradable.

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